

## 2.5. Heuristic algorithms

A number of known tour construction and tour improvement heuristic algorithms for the TSP (see, e.g., Golden and Stewart [388] or Chapter 8) can be adapted to both GTSP and E-GTSP. We next concentrate on the heuristics producing feasible E-GTSP (and hence GTSP) solutions proposed by Fischetti, Salazar and Toth [300].

As to tour construction procedures, we describe a possible adaptation of the well-known *farthest insertion* TSP procedure; *nearest insertion* and *cheapest insertion* procedures can be adapted in a similar way. For each pair of clusters  $C_h$  and  $C_k$ , let the corresponding *distance*  $d_{hk}$  be defined as  $d_{hk} := \min\{c_{ij} : i \in C_h, j \in C_k\}$ . The procedure starts by choosing the two clusters, say  $C_a$  and  $C_b$ , that are farthest from each other (with respect to distances  $d_{hk}$ ), and defines a partial tour  $T$  between the two closest nodes  $i \in C_a$  and  $j \in C_b$ . At each iteration,  $T$  is enlarged by first determining the uncovered cluster  $C_h$  farthest from the clusters currently visited by  $T$ , and then by inserting a node  $v$  of  $C_h$  between two consecutive nodes  $i$  and  $j$  of  $T$  so as to minimize  $c_{iv} + c_{vj} - c_{ij}$ . The procedure stops when  $T$  covers all the clusters. As in the TSP case, the procedure is likely to produce better solutions when the costs satisfy the triangle inequality.

We next describe two tour improvement procedures.

The first procedure, RP1, is based on 2-opt and 3-opt exchanges. Let  $T$  be the current E-GTSP solution, visiting exactly one node for each cluster, and let  $S \subseteq V$  be the set of the visited nodes. Clearly, any near-optimal TSP solution on the subgraph induced by  $S$  (found heuristically through, e.g., 2- or 3-opt exchanges) can lead to an improved GTSP solution. This approach has however the drawback of leaving the set of visited nodes unchanged. In order to remove this restriction, the following generalized 2-opt scheme has been proposed. Let  $(\dots, C_\alpha, C_\beta, \dots, C_\gamma, C_\delta, \dots)$  be the cluster sequence corresponding to the current tour  $T$ . All the edges of  $T$  not incident with the nodes in  $C_\alpha \cup C_\beta \cup C_\gamma \cup C_\delta$  are fixed. The scheme tries to exchange the current cluster sequence into  $(\dots, C_\alpha, C_\gamma, \dots, C_\beta, C_\delta, \dots)$ . To this end, two node pairs  $(u^*, w^*)$  and  $(v^*, z^*)$  are determined such that

$$\begin{aligned} c_{iu^*} + c_{u^*w^*} + c_{w^*h} &= \min\{c_{ia} + c_{ab} + c_{bh} : a \in C_\alpha, b \in C_\gamma\}, \\ c_{jv^*} + c_{v^*z^*} + c_{z^*k} &= \min\{c_{ja} + c_{ab} + c_{bk} : a \in C_\beta, b \in C_\delta\}, \end{aligned}$$

where nodes  $i, j, h$  and  $k$  are the nodes visited by  $T$  belonging to the clusters preceding  $C_\alpha$ , following  $C_\beta$ , and preceding  $C_\gamma$  and following  $C_\delta$ , respectively.

This computation requires  $|C_\alpha| |C_\gamma| + |C_\beta| |C_\delta|$  comparisons. On the whole, trying all the possible pairs  $(C_\alpha, C_\beta)$  and  $(C_\gamma, C_\delta)$  leads to an  $O(n^2)$  time complexity, since each edge of  $G$  needs to be considered only twice.

Moreover, RP1 considers a 3-opt exchange trying to modify the cluster sequence  $(\dots, C_\alpha, C_\beta, C_\gamma, \dots, C_\delta, C_\varepsilon, \dots)$  into  $(\dots, C_\alpha, C_\gamma, \dots, C_\delta, C_\beta, C_\varepsilon, \dots)$ .

We next describe a second refinement procedure, RP2, that proved to be quite effective in our computational study. Let  $T$  be the current E-GTSP solution, and let  $(C_{h_1}, \dots, C_{h_m})$  be the sequence in which  $T$  goes through the clusters. The refinement consists of finding the best feasible tour,  $T^*$ , visiting the clusters according to the given sequence. This can be done, in polynomial time, by solving  $|C_{h_1}|$  shortest path problems, as described below.

We construct a layered acyclic network,  $LN$ , having  $m + 1$  layers corresponding to clusters  $C_{h_1}, \dots, C_{h_m}, C_{h_1}$ ; see Figure 2.5, where all edges are directed from left to right.  $LN$  contains all the nodes of  $G$ , plus an extra node  $j'$  for each  $j \in C_{h_1}$ . There is an arc  $(i, j)$  for each  $i \in C_{h_t}$  and  $j \in C_{h_{t+1}}$  ( $t = 1, \dots, m - 1$ ), having cost  $c_{ij}$ . Moreover, there is an arc  $(i, j')$  for each  $i \in C_{h_m}$  and  $j \in C_{h_1}$ , having cost  $c_{ij}$  (these arcs connect the last two layers of the network). For a given  $w \in C_{h_1}$ , any path in  $LN$  from  $w$  to  $w'$  visits exactly one node for each layer (cluster), hence it gives a feasible E-GTSP tour. Conversely, every E-GTSP tour visiting the clusters according to sequence  $(C_{h_1}, \dots, C_{h_m})$  corresponds to a path in  $LN$  from a certain  $w \in C_{h_1}$  to  $w'$ . It then follows that the best E-GTSP tour  $T^*$  visiting the clusters in the same sequence, can be found by determining the shortest path from each  $w \in C_{h_1}$  to the corresponding  $w'$ . The overall time complexity is then  $|C_{h_1}| O(n^2)$ , i.e.,  $O(n^3)$  in the worst case. In practice, the time typically spent is significantly reduced by choosing  $C_{h_1}$  as the cluster with minimum cardinality, and using a shortest-path algorithm specialized for acyclic digraphs (e.g., Bang-Jensen and Gutin [84] and Cormen, Leiserson and Rivest [218]).

Notice that the above refinement procedure leads to an  $O((m-1)! n^3)$ -time exact algorithm for E-GTSP, obtained by trying all the  $(m-1)!$  possible cluster sequences. Therefore, E-GTSP is polynomially solvable for fixed  $m$  (independently of  $n$ ).

## 2.6. A branch-and-cut algorithm

In this subsection we describe the enumerative algorithm for the exact solution of the problem proposed by Fischetti, Salazar and Toth in [300]. Since all the instances considered in the computational study have trian-